

13	B	$PV = \frac{1}{3}Nm \langle c^2 \rangle$ V and m remain unchanged $\Rightarrow \frac{N \langle c^2 \rangle}{P} = \text{constant}$ $\Rightarrow \frac{N \times 3.0 \times 10^5}{P} = \frac{\frac{3N}{4} \langle c^2 \rangle}{(P/2)}$ $\Rightarrow N = 2.0 \times 10^5 \text{ m}^2 \text{ s}^2$
14	A	This is an SHM question and the E graph is A lowest at the equilibrium and maximum